

Assignment 4&5

Empirical Test (Back Test, Forward Test, Paper Trading, Live Trading)	
Benchmark	S&P 500 Index (2015–2025): ~13% annualized return, Sharpe ~0.80, Max Drawdown ~-34%. Used as passive benchmark. Strategy targets positive alpha with lower drawdown than benchmark.
Performance Assessment	Optimized parameters via grid search across signal windows (5–90 days) and entry thresholds ($k = 0.6\text{--}2.5\sigma$). No stop-loss applied. Copper → ROK: Best params = window 15d, $k = 1.5\sigma$. Training Sharpe = 0.166. Copper → TEL: Best params = window 15d, $k = 1.2\sigma$. Training Sharpe = -0.065 (all combinations negative). Copper → ROK is a viable but early-stage strategy. Across the full backtest period, ROK delivers a positive training CAGR of 5.51% and holds up out-of-sample at 2.58%, with a Profit Factor above 1.0 in five of six test periods, confirming that wins are meaningfully larger than losses even when they do not occur more often than losses. The strategy performs best in commodity-driven stress environments, producing a 16.26% CAGR and Profit Factor of 2.116 during the 2022 Rate Hike Cycle and 8.25% CAGR with a Profit Factor of 2.210 during the US-China Trade War, which are precisely the conditions the strategy is designed to exploit. The primary weaknesses are a maximum drawdown of -24.55% in training and near-zero Sharpe ratios out-of-sample (-0.005). Once stop-loss has been implemented, I expect out of sample metrics to change. Copper → TEL is not yet tradeable. TEL's training Profit Factor of 0.730 and win rate of 32.50% indicate the signal is generating more losing trades than winning ones, and the -89.84% maximum drawdown during training represents a near-total loss of capital that disqualifies it from live deployment under current parameters. The one consistent finding across both pairs is that the strategy breaks down during extreme macro dislocations such as COVID (where ROK's Profit Factor collapsed to 0.159), strongly supporting the addition of a VIX regime filter as the first priority in Phase 2 before any live trading begins.

Back Test 1	Period: 2015–2025. Contains both bull markets (2016–2019, 2020–2021 recovery) and bear conditions (2018 trade war selloff, 2020 COVID crash, 2022 rate hike drawdown). 70% Training (2015–2022) / 30% Out-of-Sample Test (2022–2025). Position size: 95% of fixed initial capital per trade. Exit: Z-score reverts to $\pm 0.25\sigma$ or 20-day max hold. Copper → ROK — Training: CAGR 5.51%, Simple Return 6.50%, Sharpe 0.166, Calmar 0.224, Max Drawdown -24.55%, Win Rate 49.06%, Profit Factor 1.717, 53 trades, Avg Hold 12.8 days, Frequency 7.6 trades/yr. Copper → ROK — Out-of-Sample: CAGR 2.58%, Simple Return 2.65%, Sharpe -0.005, Calmar 0.119, Max Drawdown -21.60%, Win Rate 45.00%, Profit Factor 1.281, 20 trades, Avg Hold 13.6 days, Frequency 6.7 trades/yr. Strategy generalizes positively out-of-sample. Copper → TEL — Training: CAGR -10.05%, Simple Return -7.48%, Sharpe -0.065, Calmar -0.112, Max Drawdown -89.84%, Win Rate 32.50%, Profit Factor 0.730, 80 trades. Signal not viable during training period. Copper → TEL — Out-of-Sample: CAGR -3.24%, Sharpe -0.353, Max Drawdown -27.42%, Win Rate 34.48%, Profit Factor 0.823, 29 trades. Losses reduced OOS but strategy remains unprofitable.
Back Test 2	Period: July 2024 – January 2025 (nearest 6 months). Tests whether the strategy remains relevant in the most recent market environment. Copper → ROK: CAGR -10.41%, Sharpe -1.026, Calmar -1.282, Max Drawdown -8.12%, Win Rate 50%, Profit Factor 1.041, 2 trades. Very few trades fired — copper was rangebound in H2 2024, suppressing signal generation. Sample too small for statistical inference. Copper → TEL: CAGR 4.81%, Sharpe 0.124, Calmar 1.120, Max Drawdown -4.30%, 1 trade, 0 wins. Single trade insufficient for conclusions. Positive Calmar may be coincidental. Note: Both pairs generated very few signals in this window. A minimum of 20+ trades is required for meaningful back test inference.

Back Test Scenario 1	Scenario: US-China Trade War (January 2018 – December 2019). A commodity-driven risk event where copper prices declined sharply due to tariff uncertainty and global growth fears. Copper → ROK: CAGR 8.25%, Sharpe 0.362, Calmar 0.601, Max Drawdown -13.73%, Win Rate 53.85%, Profit Factor 2.210, 13 trades. Best scenario for ROK where the strategy correctly identified that equities lagged the copper selloff and profited from the directional repricing. Copper → TEL: CAGR -11.91%, Sharpe -1.052, Calmar -0.450, Max Drawdown -26.49%, Win Rate 35.00%, Profit Factor 0.523, 20 trades. Worst scenario for TEL. TEL equity was influenced by idiosyncratic factors during this period that overrode the commodity signal.
Back Test Scenario 2	<u>Scenario 2a:</u> COVID Crash and Recovery (January 2020 – June 2021). Extreme dislocation where macro factors dominated all asset pricing simultaneously. Copper → ROK: CAGR -5.31%, Sharpe -0.684, Calmar -0.322, Max Drawdown -16.47%, Win Rate 28.57%, Profit Factor 0.159, 7 trades. Strategy struggled — COVID dislocations overwhelmed commodity-equity lag signals. Idiosyncratic supply chain and demand shocks drove ROK independently of copper. Copper → TEL: CAGR 2.22%, Sharpe -0.003, Calmar 0.148, Max Drawdown -15.04%, Win Rate 38.46%, Profit Factor 1.264, 13 trades. TEL marginally positive — pairs show complementary behavior across regimes. <u>Scenario 2b:</u> 2022 Federal Reserve Rate Hike Cycle (January 2022 – December 2022). Copper declined with global growth fears; industrials repriced with a lag. Copper → ROK: CAGR 16.26%, Sharpe 0.638, Calmar 0.890, Max Drawdown -18.28%, Win Rate 50.00%, Profit Factor 2.116, 6 trades. Strongest annualized return across all scenarios. Strategy well-suited to commodity-driven bear markets. Copper → TEL: CAGR 8.54%, Sharpe 0.309, Calmar 0.573, Max Drawdown -14.91%, Win Rate 57.14%, Profit Factor 1.993, 7 trades. Both pairs profitable. Best period for TEL — highest win rate across all tests. Key finding: Both pairs perform best during commodity-driven macroeconomic stress (trade war, rate hikes). Both struggle during extreme idiosyncratic dislocations (COVID).

Paper Trading (Forward Test)	Plan: Monitor copper futures (HG=F), ROK, and TEL daily after market close. Compute Z-score using optimized parameters. Log all signal triggers, entry prices, exit prices, hold durations, and realized P&L. Compare realized signals against backtest expectations to detect parameter drift. Copper → ROK: Active paper trading. Signal: Z-score (window=15d, $k=1.5\sigma$). Entry: short when $Z > +1.5\sigma$, long when $Z < -1.5\sigma$. Exit: Z reverts to $\pm 0.25\sigma$ or 20 days elapsed. Copper → TEL: Observe-only phase. Strategy not yet profitable on OOS data. Monitor signal firing frequency and directional accuracy before committing to paper trades.
Live Trading	Proceed to live trading for Copper → ROK only after paper trading validation confirms Sharpe > 0.3 over 6+ months and 15+ trades. Phase 2 additions before live deployment: (1) Add stop-loss at -2% position NAV. (2) Add VIX regime filter — no new entries when $VIX > 30$. (3) Optimize capital allocation across pairs using volatility-weighted or Kelly sizing. (4) Begin with minimum capital; scale after 20+ live trades confirm backtest replication. Copper → TEL: not recommended for live trading until out-of-sample performance turns positive across multiple test windows.
Capacity Test	Copper → ROK (Rockwell Automation): Average daily volume ~500K shares. Strategy trades approximately 7 times per year with average hold of 13.6 days. Estimated capacity without meaningful market impact: up to \$500K per position. Copper → TEL (TE Connectivity): Average daily volume ~1.2M shares. Higher liquidity than ROK. Strategy capacity estimated at \$1M–\$2M before slippage becomes a constraint. Overall strategy capacity: \$500K–\$2M across both pairs under current signal frequency (~7–10 trades per pair per year). Capacity constrained primarily by signal frequency, not liquidity. Adding more validated pairs (target: 20–50) would increase total strategy capacity proportionally.

Evaluation of Empirical Results

Refine the Strategy

Does the strategy's return, risk, capacity, and trading frequency align with expectations?

Copper → ROK broadly aligns with expectations: positive CAGR (5.51% training, 2.58% OOS), Profit Factor above 1.0 in five of six periods, and ~7 trades per year with ~13-day average holds. Copper → TEL does not align. Training Sharpe of -0.065 and Profit Factor of 0.730 indicate the signal is not working under current parameters and requires further refinement before deployment.

What caused the drawdowns, and are there methods to mitigate them?

ROK's largest drawdown (-24.55% training) occurred during the 2018 trade war and COVID period, where commodity-equity correlations broke down and idiosyncratic equity factors overwhelmed the signal. TEL's extreme drawdown (-89.84%) reflects consistent signal misalignment throughout the training period. Primary mitigation: (1) add a VIX regime filter to block new entries when VIX > 30, preventing the worst COVID-era trades; (2) introduce a position-level stop-loss of -2% NAV in Phase 2 after parameter optimization is complete.

Would earlier or later entry/exit timing improve performance?

The grid search tested signal windows from 5 to 90 days. The optimal window for both pairs was 15 days. This suggests the commodity-equity lag is tighter than originally hypothesized, meaning the market reprices faster and earlier entry (within 1–2 weeks of the commodity move) captures more of the available spread. The exit threshold of $\pm 0.25\sigma$ is working well given average holds of 12–14 days, which confirms positions are closing on mean reversion rather than timing out. Testing a tighter exit threshold (e.g., $\pm 0.15\sigma$) could reduce hold times and free capital for the next signal.

Can risk be reduced or trading frequency increased while maintaining target returns?

Risk can be reduced without sacrificing returns by adding the VIX filter and stop-loss capabilities. Trading frequency is currently 6–8 trades per year for ROK, low but appropriate given the signal quality. Frequency could be increased by lowering k_sigma , but the grid search shows this consistently reduces Sharpe ratios, meaning the additional trades add noise rather than signal. A better path to increased frequency is expanding the pair universe: adding 5–10 validated copper-industrial pairs, or other pairs would multiply signal opportunities without degrading per-pair quality.

Can the strategy be extended to other instruments or markets to expand opportunities?

Yes. The framework is commodity-agnostic and has already been tested across copper, crude oil, natural gas, steel, and lumber. The next validated expansion candidates are Copper → Hubbell (HUBB) and Copper → Freeport-McMoRan (FCX, producer-side pair), both pending statistical validation. Longer term, the strategy can be extended to: (1) international equities with high commodity input dependency, (2) ETF-level pairs (e.g., copper ETF vs. industrial sector ETF) for higher capacity, and

	<i>(3) seasonal commodity pairs such as natural gas vs. utility stocks where seasonal demand cycles create predictable lag patterns.</i>
Applicable Market Conditions	The strategy performs best in commodity-driven macroeconomic stress environments where copper prices move sharply and industrial equities reprice with a lag. The two strongest observed scenarios were: <i>2022 Federal Reserve Rate Hike Cycle: ROK CAGR 16.26%, Sharpe 0.638, Profit Factor 2.116. Copper declined with global growth fears while industrial equities were slow to reprice downward — exactly the information diffusion lag the strategy exploits.</i> <i>US-China Trade War (2018–2019): ROK CAGR 8.25%, Sharpe 0.362, Profit Factor 2.210. Tariff uncertainty drove copper lower while ROK's equity price lagged the repricing, generating clean directional signals.</i> The strategy performs poorly in extreme macro dislocations where correlations across all asset classes break down simultaneously. COVID (2020) is the primary example: ROK Profit Factor collapsed to 0.159 as supply chain disruptions and demand shocks drove equity prices independently of commodity signals. High VIX environments ($VIX > 30$) are the measurable indicator of these regime shifts. In summary: the strategy is best suited to periods of elevated but not extreme volatility, directional commodity trends (not rangebound markets), and macroeconomic environments where input cost pressures are the dominant driver of industrial equity earnings. It is not suited to sudden systemic dislocations, rangebound commodity markets (as seen in H2 2024), or periods where company-specific news dominates sector-wide commodity signals.

Risk / Position Management	Position Control: Each trade deploys 95% of a fixed initial capital allocation per pair. This is a fixed-fractional approach that prevents exponential compounding of losses. Capital allocated per pair is determined by volatility-weighted sizing across the pair portfolio where pairs with lower historical drawdown receive proportionally more capital. Stop-Loss (Phase 2): No stop-loss is applied in Phase 1 to avoid overfitting parameters before the base strategy is validated. In Phase 2, a position-level stop-loss of -2% NAV will be introduced. This is calibrated to be wider than normal daily equity volatility (~0.8–1.2% for large-cap industrials) to avoid premature exits, while capping the maximum loss per trade at a manageable level. Stop-Gain: No explicit stop-gain is applied. The mean-reversion exit (Z-score returning to $\pm 0.25\sigma$) functions as a dynamic profit target: the position closes when the mispricing has closed, regardless of the dollar amount gained. A fixed stop-gain would risk exiting prematurely before the full spread has closed. Hedging: The strategy is equity-only (no commodity leg). This is an intentional simplification for Phase 1. In Phase 2, a partial hedge using commodity ETF options could be introduced to reduce exposure to commodity reversals that occur after entry but before the equity reprices. Risk Red Flags: <i>Liquidity risk: Both ROK and TEL are large-cap with sufficient daily volume at current position sizes. Becomes relevant above \$2M per position.</i> <i>Systemic risk: Broad market dislocations ($VIX > 30$) break commodity-equity correlations. Addressed by VIX regime filter in Phase 2.</i> <i>Model risk: The cointegration relationship between copper and ROK/TEL may weaken over time as business models evolve (e.g., ROK increasing software revenue). Rolling beta re-estimation (252-day window) partially mitigates this but does not eliminate it. Cointegration tests should be re-run annually.</i> <i>Concentration risk: Both current pairs are copper-industrial. A single copper market disruption (e.g., major mine strike, Chinese demand collapse) could trigger correlated losses across both pairs simultaneously. Expanding to other commodity categories in Phase 2 reduces this.</i> <i>Earnings event risk: Large quarterly earnings surprises can overwhelm the commodity signal. Mitigated by the existing rule blocking new entries within 5 days of earnings announcements.</i>
-----------------------------------	---

Execution Risk Assessment	Order Fill Rate: The strategy trades large-cap US equities with high daily liquidity. At current position sizes (retail to small institutional), order fill rates will be effectively 100% at market open or close. Fill rate becomes a concern only above \$1–2M per order, where limit orders rather than market orders would be required to avoid price impact. Execution Latency: This is a low-frequency strategy (6–8 trades per year, signals computed after daily close). Execution latency of minutes to hours is immaterial — the signal window is 15 days, so a same-day or next-morning execution introduces negligible timing error. Latency is not a meaningful risk at this frequency. Slippage: Estimated at 0.10–0.20% per trade for both equities at current position sizes, based on bid-ask spreads of \$0.01–0.05 for large-cap US equities. At 7 trades per year, annual slippage cost is approximately 0.70–1.40% of deployed capital, already partially accounted for in the 0.10% per-trade commission applied in the backtest. At larger position sizes, slippage scales with the square root of participation rate relative to average daily volume. Market Impact: Negligible at retail and small institutional scale. ROK average daily volume ~\$140M; TEL ~\$156M. A \$100,000 position represents 0.07% of daily volume for ROK and 0.06% for TEL which is well below the 1% threshold where market impact becomes measurable. Impact becomes material above \$1.4M (ROK) and \$1.56M (TEL) per trade. Leverage: No leverage is used in Phase 1. All positions are fully cash-funded. Introducing leverage in a later phase would amplify both returns and drawdowns proportionally, given the current maximum drawdown of -24.55% for ROK, even 2x leverage would produce drawdowns approaching -50%, which is likely beyond acceptable risk tolerance for most investors. Capital Gains Tax: The strategy's average holding period of 12–14 days means all realized gains are classified as short-term capital gains, taxed at ordinary income rates (up to 37% in the US) rather than the preferential long-term rate (20%). This is a meaningful drag on after-tax returns. A position held for just 6 additional days beyond the current average would cross the 20-day maximum hold limit. This tax drag should be explicitly modeled when reporting net returns to investors.
----------------------------------	--

Below are the results of the statistical tests for the 19 pairs I tried.

#	Pair	Commodity	Equity	Coint p-value	Coint Pass	Granger p-value	Granger Lag (days)	Granger Pass	Beta (β)	R ²	Both Tests Pass
1	Copper → TE Connectivity	HG=F	TEL	0.0011	✓	0.0407	5	✓	0.43	0.13	✓✓ TRADE
2	Copper → Rockwell Automation	HG=F	ROK	0.0200	✓	0.0161	8	✓	0.47	0.13	✓✓ TRADE
3	Copper → Ametek	HG=F	AME	0.2997	✗	0.0024	5	✓	0.35	0.11	✗ SKIP
4	Copper → Parker Hannifin	HG=F	PH	0.9563	✗	0.0184	5	✓	0.48	0.12	✗ SKIP
5	Copper → Eaton Corp	HG=F	ETN	0.9503	✗	0.0103	5	✓	0.41	0.11	✗ SKIP
6	Copper → Ingersoll Rand	HG=F	IR	0.8216	✗	0.0154	5	✓	0.45	0.08	✗ SKIP
7	Copper → GE Vernova	HG=F	GEV	0.9755	✗	0.0140	6	✓	0.39	0.06	✗ SKIP
8	Copper → Emerson Electric	HG=F	EMR	0.2290	✗	0.2888	7	✗	0.44	0.14	✗ SKIP
9	Copper → Aptiv	HG=F	APTV	0.6608	✗	0.6577	5	✗	0.53	0.09	✗ SKIP
10	Crude Oil → American Airlines	CL=F	AAL	0.3628	✗	0.0000	10	✓	0.04	0.01	✗ SKIP
11	Crude Oil →	CL=F	UAL	0.6171	✗	0.0001	10	✓	0.04	0.01	✗ SKIP

	United Airlines										
12	Crude Oil → Delta Airlines	CL=F	DAL	0.2083	x	0.0000	10	✓	0.03	0.00	x SKIP
13	Natural Gas → CF Industries	NG=F	CF	0.7717	x	0.0079	1	✓	0.09	0.02	x SKIP
14	Natural Gas → Nutrien	NG=F	NTR	0.1199	x	0.3007	3	x	0.07	0.02	x SKIP
15	Steel → Ford	SLX	F	0.0615	x	—	—	x	0.55	0.26	x SKIP
16	Steel → General Motors	SLX	GM	0.2806	x	—	—	x	0.58	0.28	x SKIP
17	Lumber → Lennar	LBS=F	LEN	0.9467	x	0.2380	8	x	0.15	0.04	x SKIP
18	Lumber → D.R. Horton	LBS=F	DHI	0.9452	x	0.1393	2	x	0.14	0.04	x SKIP
19	Lumber → PulteGroup	LBS=F	PHM	0.9887	x	0.1240	2	x	0.17	0.05	x SKIP

Here is a link to my github where I perform the statistical test to evaluate pairs and evaluate the strategy using backtest. The key python scripts are test_pairs.py and backtest_fixed.py.

<https://github.com/jahnavi-maddhuri/commodity-equity>

AI Acknowledgement: I used Claude 4.6 for this assignment. All questions and answers are in this chat (continued from previous assignments).

<https://claude.ai/chat/7e1d6cd3-ddba-48a0-9c79-286fd770d2cc>